

Lwazi Knowledge Gumede

Systems Engineer (Engineer-in-Training) | ECSA Candidate Engineer | BSc (Honours) Computer Engineering

lwazig28@gmail.com | +27 76 935 2103 / +27 65 711 1226 | Johannesburg, home in Umlazi, Durban. Willing to relocate.



Professional summary

ECSA-registered Candidate Engineer working as a systems engineer at Transnet Rail Infrastructure Manager, building and shipping full-stack software inside a live rail-infrastructure environment. Combines hands-on field engineering, including CAS testing, commissioning, RailBAM calibration and derailment investigations, with production software delivery in Python, Django, React, TypeScript, PHP and PostgreSQL. Strong in system automation, data engineering and REST API development, with a consistent record of replacing manual workflows with reliable automated tools. Comfortable across hardware and software through embedded and IoT systems, and doing MSc research at North-West University on machine-learning fault localisation in photovoltaic arrays.

Work experience

Systems Engineer (Engineer-in-Training)

April 2025 – Present

Transnet SOC Ltd (Rail Infrastructure Manager) | Johannesburg

- Replaced manually compiled spreadsheets, saving the team the equivalent of roughly a week of manual work each month, by designing and deploying full-stack monitoring dashboards in React, TypeScript, Python/Django, PostgreSQL and WebSocket that gave the Technical Office and corridor depots a single live view of rail operations.
- Removed repetitive manual logging from daily engineering workflows by building automated Python/REST backend services for data collection, vehicle identification and compliance reporting.
- Strengthened asset-reliability and safety decisions by running technical reviews, field testing and commissioning of Condition Assessment Systems (CAS) and supporting derailment investigations with documented analysis and configuration management.
- Owns R&D engineering change proposals (ECPs) and the procurement workflows that follow them, end to end across departments, from technical motivation through to award.
- Stayed on track for professional registration by logging engineering competencies against ECSA candidate requirements in a documented professional-development logbook.

Key projects

- **Rail Monitor.** Live equipment availability across points, signals, TVDS, power and remote control, scored with a weighting per equipment type from SAP, GPS and operator input validated weekly with the Technical Office. Designed and built it, including the back-end logic for substation breaker and switching states, and presented the phase 2 plan (station level to asset level) to stakeholders. React, TypeScript, Django, PostgreSQL, WebSocket.
- **Train movement graph optimisation.** Rewrote legacy sequencing logic so each CTC section gets its own graph, and added a historical archive beside the real-time database, a caching layer, a background export worker and a live versus historical overlay. Processed large date ranges in chunks to avoid memory exhaustion. PHP, PostgreSQL, Highcharts.
- **Trunking highsight dashboard.** Consolidated the Network Operations Centre's separate fault screens into one view with long-outstanding fault tracking and asset visualisation, traced silently failing ingestion scripts, and moved data collection onto background services. React, Python, PostgreSQL, Windows services.
- **CAS availability dashboard.** Rebuilding the CAS dashboard by scraping legacy data into PostgreSQL, and moving new development into production as one system with a per-table impact list and a test for each migration. PostgreSQL, PHP.
- **VIS Master.** Health monitoring, communication diagnostics and status tracking for Vehicle Identification Systems across the rail network.
- **RailBAM calibration.** Calibrated trackside acoustic cabinets and wayside hut hardware for bearing and wheel defect detection (RS, LF, WHLFLT), set speaker set-up points and sensitivity thresholds, and confirmed sensor performance in the trending database afterwards.
- **Offline ML platform for train movement data (in development).** Designed a fully offline, CPU-only platform covering delay prediction (LightGBM), demand forecasting (Prophet, LSTM), schedule optimisation (OR-Tools CP-SAT), anomaly detection (Isolation Forest, autoencoder) and position prediction (Kalman filter, XGBoost). FastAPI, Streamlit, MLflow, DuckDB.

Engineering Tutor & Laboratory Demonstrator

February 2024 – May 2025

University of KwaZulu-Natal | 5 modules, first- and second-year engineering

- Raised the standard of student technical reports and presentations by running Technical Communication (ENCH1TC) tutorials and giving structured written feedback, earning consistently positive lecturer and student ratings for clarity.
- Helped students turn theory into working solutions by demonstrating Computer Methods (programming and numerical analysis), Electronic Design, Electrical Principles (ENEL2EB) and Field Theory (ENEL2FT) labs and guiding projects from concept to implementation.
- Ensured accurate experimental results and full safety compliance by setting up, verifying and supervising practical lab equipment for second-year cohorts.

Vacation Student (Engineering Internship)

January 2023 – March 2023

Transnet Engineering | Durban

- Gave engineering stakeholders self-serve access to operational data by building Power BI dashboards that turned raw engineering metrics into clear visual reports for decision-making.
- Eliminated manual data hand-offs by implementing Python-scripted pipelines that automatically pushed data into shared Google Sheets for real-time, cross-department access.
- Built early exposure to rail operations, engineering project management and safety protocols within a large state-owned enterprise.

Research

MSc, Computer & Electronic Engineering

2026 – Present

North-West University | Research in progress

Large-sparse historical consistent neural networks for spatially resolved fault detection and localisation in photovoltaic arrays. The sparsity mask is derived from the array's electrical wiring graph, the model trains on healthy operating data only, and faults are located from the spatial geometry of the prediction-error residuals. Validation runs against a component-level Simscape Electrical array model, built programmatically in MATLAB. MATLAB, Simulink, Simscape Electrical.

Education

BSc (Honours) in Computer Engineering

2019 – June 2024

University of KwaZulu-Natal | ECSA-accredited

Focus areas: embedded systems, IoT, software development, digital signal processing, control systems, data structures and algorithms, database systems. Key projects: IoT smart meter (ESP32, PHP, MySQL), blockchain invoice discounting platform (Solidity, Web3.js, React), authentication and key management system, PIC16 light chaser reaction game in assembly, wavelet and Huffman image compression, art gallery intruder alarm (C++, UML), temperature-controlled fan motor drive, largest rectangular area search, digital sequence counter, Java GUI calculator.

National Senior Certificate (NSC / Matric)

2018

Nombuso High School | Bachelor's Degree Pass

Skills

Programming languages	Python, TypeScript, JavaScript, C++, C, Java, C#, PHP, SQL, Solidity, Bash, PowerShell, R, Assembly, MATLAB
Web development	React, Django, Flask, FastAPI, Node.js, Next.js, Express.js, HTML5, CSS3, RESTful APIs, WebSocket, Highcharts, D3.js, Chart.js
Data and ML	Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, LightGBM, XGBoost, Prophet, OR-Tools, MLflow, Streamlit, Power BI
Databases	PostgreSQL, MySQL, SQLite, Oracle DB, MongoDB, DuckDB, database design and optimisation
Tools and platforms	Git. GitHub. GitLab. Docker. Linux. Windows services (NSSM). VS Code. Postman
Languages: English, Zulu, Xhosa	
ECSA Candidate Engineer Transnet EIT Programme, Apr 2025-Mar 2027	

REFERENCES

Lungelihle Jafta - Line Manager, Transnet RIM Lungelihle.Jafta@transnet.net	Dr. Jules-Raymond Tapamo - Lecturer, Computer Methods tapamoj@ukzn.ac.za
Wayne Nelson - Lecturer, Technical Communication nelsonw@ukzn.ac.za	Adv. Dr. Ernest Bhero - Lecturer, E.E. Design bhero@ukzn.ac.za
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